Lesson 2.9 - How to Make a Frequency Polygon

> **Video:** How To Make a Frequency Polygon by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

A **frequency polygon** is a line graph of a frequency distribution. Plot the frequency of each class at the class's **midpoint**, then connect the points with straight lines. Add one extra point **at frequency 0** at each end, so the shape closes onto the x-axis.

Steps

1. Make a **frequency distribution table** with equal-width classes.
2. Find each class **midpoint**: $\text{midpoint} = \dfrac{\text{lower bound} + \text{upper bound}}{2}$
3. Add one **extra midpoint below** the first (first − class width) and one **above** the last (last + class width), each with frequency **0**.
4. Plot (midpoint, frequency) and connect the dots with straight lines.

Example - incomes in a town (class width $20k)

Class boundaries. Each class includes its lower bound but **not** its upper bound, so an income of exactly $40k goes in the 40k-60k class:

$$20 \le I < 40, \quad 40 \le I < 60, \quad \dots$$

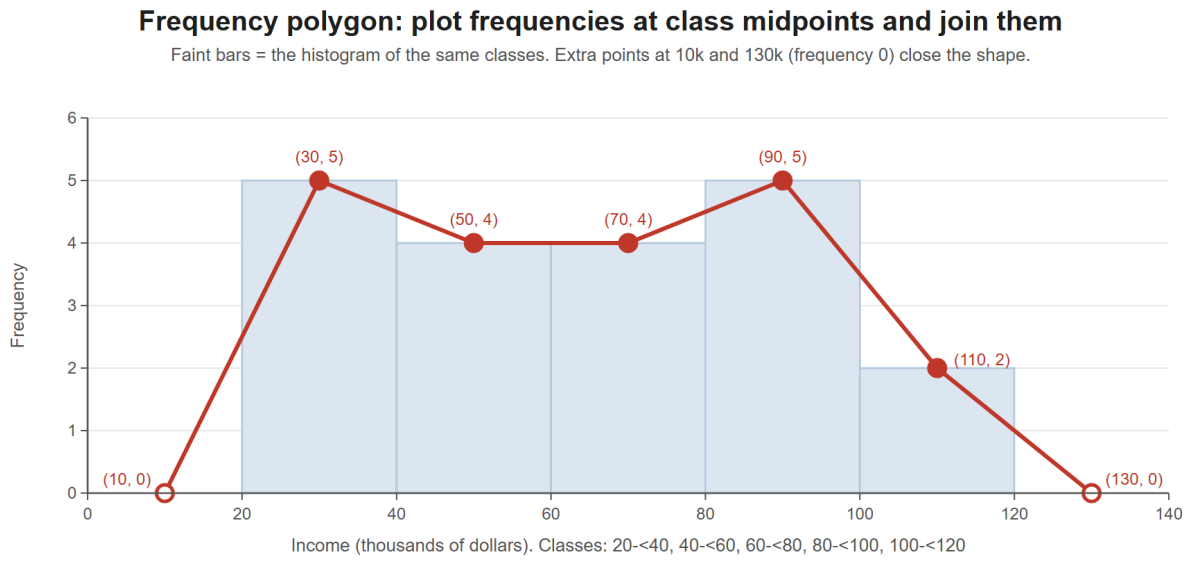
No income is below $20k, and the highest is $115k, so the classes run from 20k to 120k.

Income range (k = $1,000)	Frequency	Midpoint
20k - < 40k	5	30k
40k - < 60k	4	50k
60k - < 80k	4	70k
80k - < 100k	5	90k
100k - < 120k	2	110k
Total	**20**	

For example, the 20k-40k class contains 32k, 25k, 39k, 26k and 25k, so its frequency is 5, and its midpoint is $(20 + 40)/2 = 30$k.

Extra end points: $30 - 20 = 10$k and $110 + 20 = 130$k, both with frequency 0.

Midpoint	10k	30k	50k	70k	90k	110k	130k
Frequency	0	5	4	4	5	2	0



Each point sits at a class **midpoint**, at the height of that class's bar. The two hollow points at 10k and 130k have frequency 0 and bring the polygon down to the axis at both ends.

Key takeaways

- Plot frequencies at **class midpoints**, not at class edges.
- Add one point with **frequency 0** on each side (one class width beyond the first and last midpoints) so the polygon starts and ends on the x-axis.
- State clearly which class a boundary value such as 40k belongs to. A common convention is $[\text{lower}, \text{upper})$.